Applying effective environmental forecasts to Management Strategy Evaluation of yellowtail flounder

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1 ABSTRACT

The US Northeast Continental shelf is the fastest warming marine ecosystem in the United States, so understanding this change is imperative to maintaining the sustainability and productivity of this region's fishery. Fished stocks in the region have exhibited different distributional shifts in response to changing ocean temperatures, chemistry, and currents. Management Strategy Evaluation (MSE) is a technique in fisheries management to simulate and compare the likely outcomes of different management actions, assess trade-offs of different management options, and has been used to create more climate-ready management plans. The Georges Bank Yellowtail Flounder (*Limanda ferruginea*) is an important stock to the Northeast groundfish fishery and its recruitment has been shown to be impacted by bottom ocean temperatures. However, predicting future environmental covariates such as bottom temperature can be challenging, as these conditions are subject to high stochasticity. For this reason, our goal is to analyze the different choices made when projecting or forecasting future environmental conditions within assessments, what subsequent catch advice is generated from the MSE, and how this affects long-term fishery dynamics. Preliminary results focusing on recruitment-environment relationships have shown that different projection decisions, including integrating environmental forecasts, have a limited impact on the model performance and output, and we will expand on these findings by comparing the MSE results under different climate scenarios, informed by ocean model forecast products.

2 INTRODUCTION

As the world's oceans are complex physical and chemical systems, climate change has impacted oceans in a variety of ways such as increased water temperature and acidification, and changes in ocean currents (Fabry et al., 2008; Guinotte & Fabry, 2008; Abraham et al., 2013; Sydeman et al., 2015; Wilson et al., 2016; Poloczanska et al., 2016; García Molinos et al., 2017; Friedland et al., 2022; Cheng et al., 2024). As a result, different marine species are projected to have complex and varying responses to these environmental shifts depending on species' life histories, thermal tolerances, mobility, and other aspects of the species' biology (Guinotte & Fabry, 2008; Chown et al., 2010; Hönisch et al., 2012; Punt et al., 2014b; Stewart et al., 2014a; Sydeman et al., 2015; Poloczanska et al., 2016; Wilson et al., 2016; Chaikin et al., 2024). This creates compounding challenges when managing commercially-important fisheries. As some species will shift ranges northward or to deeper waters in search of colder temperatures, this will disrupt historical food chains as predators and prey may shift in different directions, or range shifts will introduce new interactions between predator and prey species that have historically not overlapped (Overholtz et al., 2011; Albouy et al., 2014; Stewart et al., 2014b; Sydeman et al., 2015; Wilson et al., 2016; García Molinos et al., 2017; Chaikin et al., 2024). Further, as fish populations may experience distribution shifts, annual growth patterns, or vary in abundance, this has potential to increase catch effort as historical fishing grounds and times may now not line up with fish stocks' new life history dynamics (Overholtz et al., 2011; Poloczanska et al., 2016; Tommasi et al., 2017; Fulton, 2021). New species may also shift their ranges into a commonly fished area, potentially opening up new commercial opportunities (Stewart et al., 2014a).

The US Northeast continental shelf has warmed faster than any other marine ecosystem in the United States (V. S. Saba et al., 2016; Kleisner et al., 2017; V. Saba et al., 2023). The varying responses to climate change across fished species has presented challenges in accurately assessing stock health of various species in the region, however these environmental covariates have rarely been incorporated into stock assessments in the Northeast U.S. (Hare et al., 2016; Kleisner et al., 2017; Mazur et al., 2023). These covariates have been shown to reduce retrospective patterns and increase projection accuracy in New England populations such as Atlantic Cod, haddock, yellowtail flounder, and winter flounder (Buckley et al., 2004; Klein et al., 2017). However, incorporating environmental covariates into stock assessments is difficult and a very clear and strong relationship between environmental change and population dynamics must be shown or else improper use of an environmental covariate could lead to poorer rather than better estimation performance (Haltuch & Punt, 2011a; Punt et al., 2014b; Ehrlén & Morris, 2015; Britten et al., n.d.; Miller et al., 2023a).

Management Strategy Evaluations (MSEs) are decision making frameworks that allow assessors to simulate and compare the sustainability and tradeoffs associated with different management decisions (Hollowed et al., 2011; Tommasi et al., 2017; Mazur et al., 2023; V. Saba et al., 2023). Applying climate change covariates and MSE to New England groundfish has shown to improve the robustness of stock assessments (Mazur et al., 2023). However, applications of MSEs can still be challenging, as climate change shifts population baselines and creates dynamics that are harder to predict in biological systems. This demonstrates a need for further research on how to incorporate these environmental conditions into the MSE framework (Hollowed et al., 2011; Punt et al., 2016; Stram & Holsman, 2022a; V. Saba et al., 2023). Populations' responses to climate change can be complex, as fish exhibit different sensitivities to environmental conditions depending on their lifestage, and different environmental changes (i.e. increased sea surface or bottom temperature, changes in salinity, and altered currents) will have different effects on populations (Hollowed et al., 2011).

Further, decisions on how to project environmental conditions are important to consider when predicting stock dynamics. With recent improvements, ocean dynamic models are able to offer improved predictions of future stock dynamics and are being used in stocks worldwide. However, the spatial and temporal scale of these models is relevant (Hobday et al., 2016). Future climate is also difficult to estimate as there is a lot of uncertainty and climate change is a very chaotic process (Punt et al., 2014b). As assessments attempt to capture future population trends, forecasting environmental trends is also imperative, however this can be challenging as optimal decisions can change depending on scale, species, and which environmental covariates are used (Hollowed et al., 2011; Tommasi et al., 2017; Stram & Holsman, 2022b). In a simulation-estimation experiment, Britten et al. (n.d.) found that there was no discernable difference if projections continued the AR1 process or if the environmental covariate projection was calculated as the mean of the last 5 years. This project aims to build off of this finding by expanding the projections tested, simulating over a greater number of years, and seeing how catch advice and fishery dynamics may respond to changing projections.

The yellowtail flounder (*Limanda ferruginea*) has a long history of harvest in the US Atlantic Coast. However due to stock collapse in the 1980s, it is now primarily caught as bycatch in other surrounding fisheries (CITE RTA). Yellowtail flounder is managed as three distinct stocks in the Northeast US; Southern New England, Gulf of Maine, and Georges Bank. Yellowtail flounder are sensitive to water temperatures at the larval phase (Laurence & Howell, 1981). The Georges Bank stock has been shown to shift its distribution northward in response to warming ocean temperatures (Murawski, 1993; Nye et al., 2009; Hyun et al., 2014). This stock has been in decline starting in the 1970s, leading to a collapse in the early 1990s (Stone et al., 2004). Recently, ocean bottom temperature has been shown to have a strong link to the recruitment of Georges Bank yellowtail flounder, and the incorporation of this environmental covariate has been shown to improve stock assessment accuracy and prediction success [Johnson et al. (1999); Kerr et al. (2022); CITE RESEARCH TRACK]. Because this relationship between environment and stock dynamics has a clear linkage, inclusion of environmental covariates have improved assessment success, and this stock has a long history of data collection, we will use Georges Bank yellowtail flounder as a case study. We aim to incorporate bottom temperature into stock assessments and the Management Strategy Evaluation framework in order to assess different choices when projecting bottom temperature in Georges Bank and how this affects forecasts in assessments and subsequent tactical fishery management advice of Georges Bank yellowtail flounder.

3 METHODS

To conduct the MSE process, we used trawl survey data from the Northeast Fisheries Science Center (NEFSC) and outputs from the Woods Hole Assessment Model (WHAM). WHAM was developed by researchers at NEFSC, and is a state-space stock assessment model which incorporates climate variables into fishery processes (Stock & Miller, 2021). The stock data used was collected by the NEFSC fall and spring surveys, the DFO spring survey, as well as commercial data, which spanned from 1973 to 2022. Choices for weights-at-age, natural mortality, fleet selectivity, recruitment and maturity were derived from results from the 2024 research track assessment on yellowtail flounder (CITE ASSESSMENT, see carvalho et al 2017 from class reading for good presentation of wham choices) and these numbers are available in the supplemental material. The output of the WHAM model was then used in the R package whamMSE (CITE CHENG) to simulate the MSE process. This package creates an Operating Model (OM) which simulates fishery dynamics, including fishing and stock dynamics. As a state space model, the OM incorporates process-error in its dynamics, but the values generated by the OM represent the 'true' values of the fishery. The Estimation Model (EM) simulates the survey and assessment process in the fishery, including observation errors. Stock assessments were tested with the 0.75 Fmsy NEFMC control rule. We tested a 50 year MSE period where assessments were conducted every three years.

For the environmental covariate of bottom temperature, we used a dataset developed by Du Pontavice et al. (2023), which is also used in the Georges Bank yellowtail flounder stock assessment CITE ASSESSMENT. We projected the 'real-world- temperature by conducting a linear regression on the existing data points and then projecting future temperatures along this trendline with added variation around a normal distribution. The package whamMSE was used to conduct the Management Strategy Evaluation CITECHENG. In this simulation, the environmental covariate was treated as a latent variable in the operating model, therefore process error of the ecov was turned off in the OM so only the specified projection was modeled in the "real world" scenario of our simulation. First, the Wood's Hole Assessment Model was fitted to real Yellowtail flounder data from the NEFSC survey to serve as the basis for the OM. This base-OM was used to estimate parameters for numbers-at-age, selectivity, recruitment, and ecological covariate. These parameters were then treated as 'truth' in our simulation and were used in the second OM we developed for the projections. This second OM, with all of the parameters already estimated, fixes the environmental covariate at a true state with no process error, however observation error is still included in this process. This allowed us to run multiple simulations of various temperature projections in our Estimation Model while maintaining the same 'real-world' state.

We then tested various EMs that were identical except for this temperature projection. Instead of using a single time series throughout the projection, all EMs were changed to simulate real-life stock assessment processes, where the three-year projection was recalculated at each assessment year step given the previous three years of data. Descriptions of temperature projections are described in figure 1. For the first experiment, we first compared EM temperature projections with varying levels of bias. We developed four scenarios: High Ecov, where the overall trend of the OM temperature projection was doubled to simulate a high bias in projections; Medium Ecov, where the EM uses the same temperature as the OM; Medium-Low Ecov, where the overall trend was multiplied by -1 to simulate low bias in projections; and Low Ecov, where the overall trend was multiplied by -2 to simulate a very low bias in projections. Each simulation was run 100 times to ensure robustness of model results. Any non-converged model resulted in the elimation of that whole simulation.

This simulation was then repeated with the OM changing it's baseline projection to use the High, Medium, Medium-Low, and Low temperature projections to compare how hypothetical future climate scenarios may change the outcome of having a biased projection. Each of these OMs were tested against each climate projection in the EMs.

After comparing projections through the MSE process, the experiment was repeated except

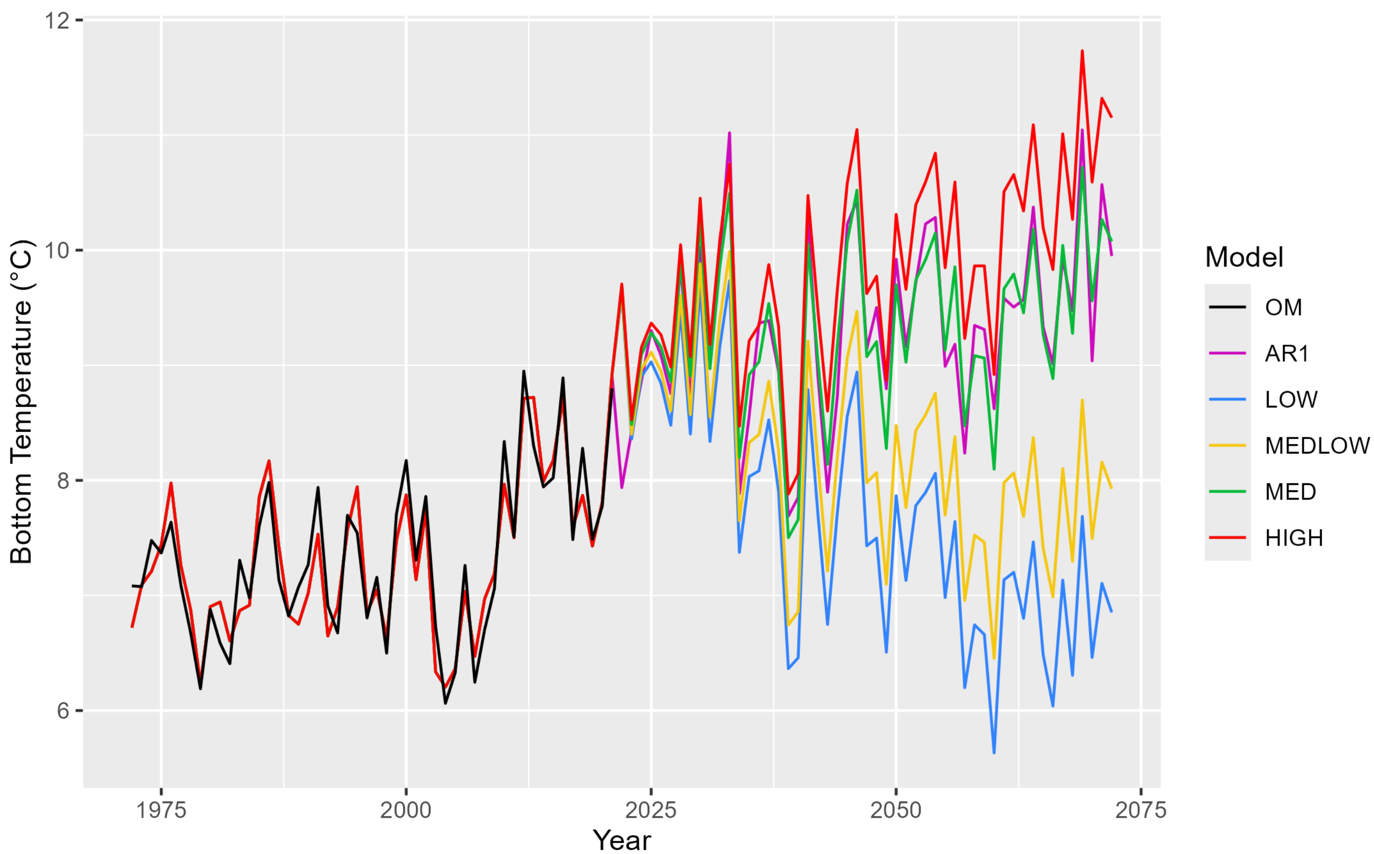


Figure 1: Temperature projections of the different Bias tests. The medium projection follows the overall trend of historical data, and this was used as a baseline for the other bias projections

we used only the original OM temperature projection (the one that maintains the trend from the overall data set). For the EMs, the two extreme bias projections were kept (High Ecov and Low Ecov), along with various projection choices that, at each assessment year, self-update with the previous years' temperature information. A new temperature projection is then calculated from that in the EM and projected for the next three years until the next assessment. The different projection choices used in the experiment are described in figure 2.

After the MSE process was conducted in both experiments, we compared the outputs of each model and the subsequent catch advice to emerge from the model. We then compared estimated natural and fishing mortality, spawning stock biomass, and estimated recruitment in each model in order to understand how different assumptions in creating environmental projections propagates reflects real-world dynamics and how this ultimately affects management decisions.

4 RESULTS

4.1 Bias Tests

The estimated parameters for the environmental link are shown in figure 3 compared to that of the OM in each bias test, and the estimated logit rho is shown in table 1. Here, we see estimated parameters ranging from 7.449 - 8.904. Generally, the EM tended to underestimate the environmental covariate compared to the OM, with the exception of High EM and the AR1 EM compared with the MediumLow OM With the exception of the MediumLowEcov, the highest logit rho (or the models whose variation was most explained by the model) for the OMs using both the highest and lowest biased projections were the EM with the highest bias, HighEcov. However, when looking at how the ecov is updated with every assessment step in the mode, the updated parameter estimates do not diverge from the 'true' estimate

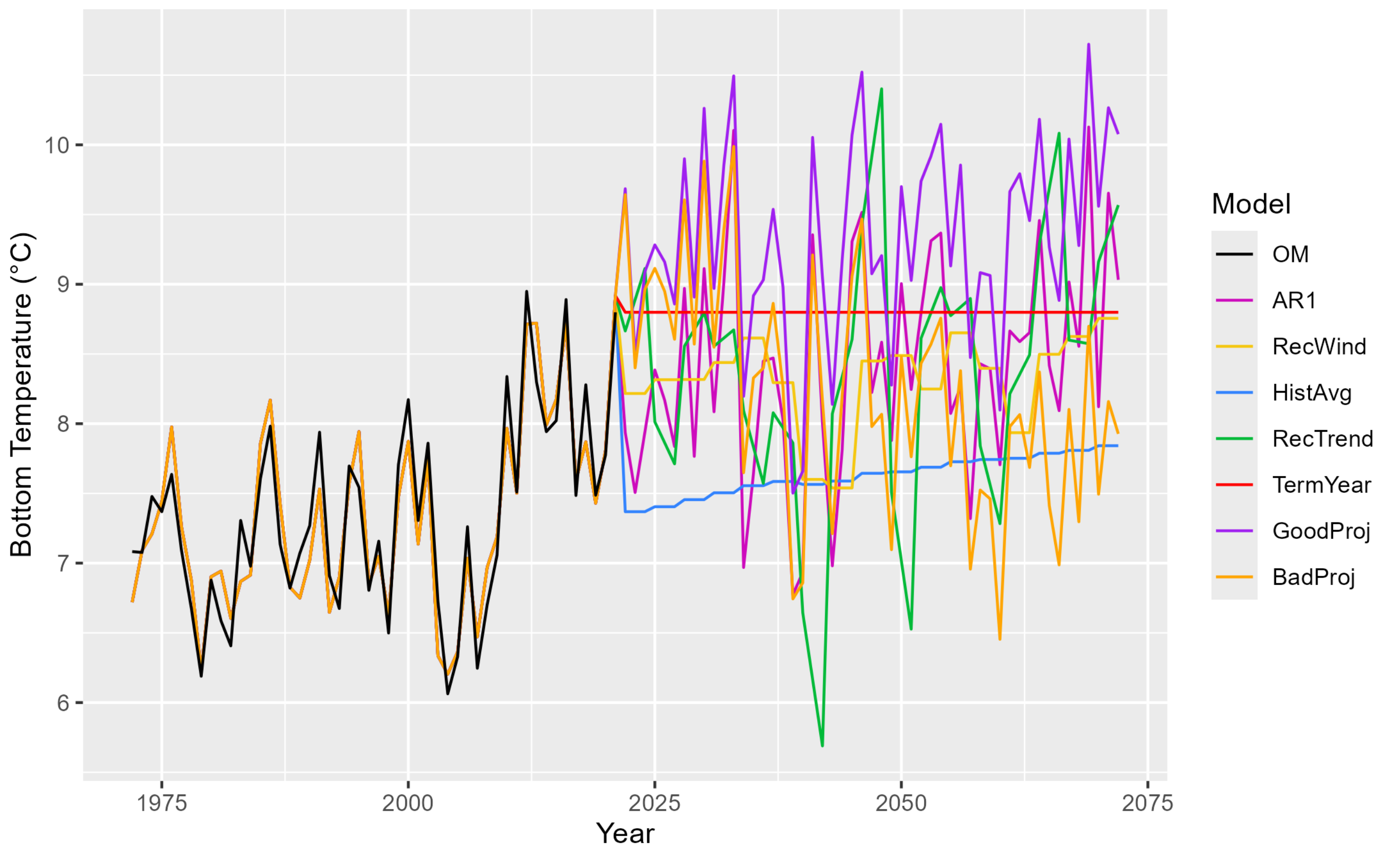


Figure 2: Temperature projections used in each EM. AR1 (Auto-Regressive1) - This projection uses an updated mean at each timestep for the projection. NewWind (New Window) - This process uses the mean of the previous 5 years in its projection. HistAvg (Historical average) - Uses the average of all datapoints collected. NewTrend (New Trend) - Similar to new window, this process takes the last 5 years of data and uses the linear regression of the last 5 years as its projection. TermYear (Terminal Year) - Uses the terminal year of historical data across the projection period. GoodProj (Good projection) - Uses the same time series as the OM. BadProj (Bad projection) - Same as the MedLow bias in the previous experiment, this projection tests projection with a low bias. NoTemp - (No temperature, not pictured) This projection does not consider environmental covariates it its projection.

in the OM, even though the underlying temperatures take different trajectories. As a result, the subsequent catch advice from the EMs, spawning stock biomass (SSB), fishing mortality ($\bar{F}$), and predicted catch from the OM stays relatively consistent across EMs over the time series. These figures are provided in the supplemental material.

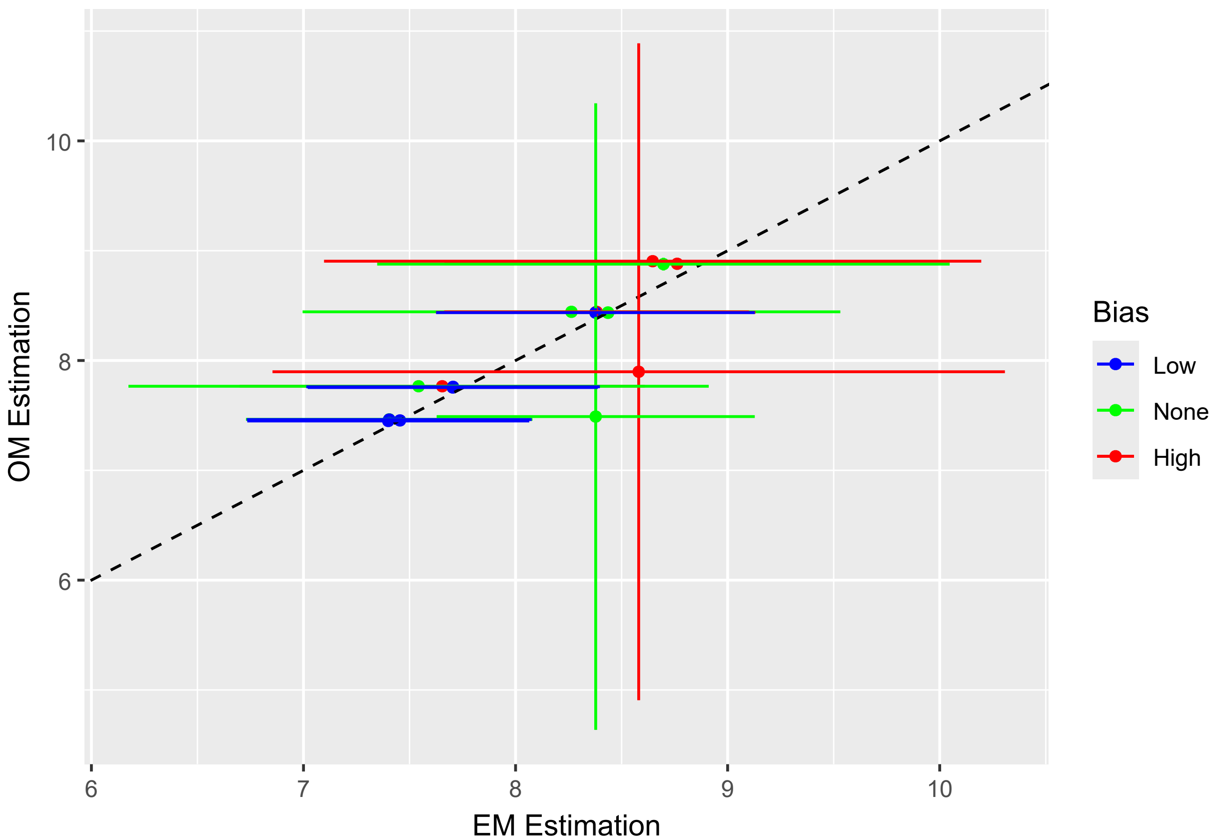


Figure 3: Estimated environmental parameter from each EM compared to the 'true' value in the OM on the y axis. Colors correspond to if the EM had a high temperature bias (red), a low temperature bias (blue), or if the EM intended to accurately reflect climate dynamics (i.e. AR1 process or temperature projection that matched that of the OM) (green). The dashed black line indicates where these two estimations would be equal

4.2 Projection Tests

When comparing the parameter for the environmental covariate across different projections (table 2), the highest parameter estimate came from the 'Good' projection which follows the general temperature trend of the OM. This model also had the highest logit rho value. The

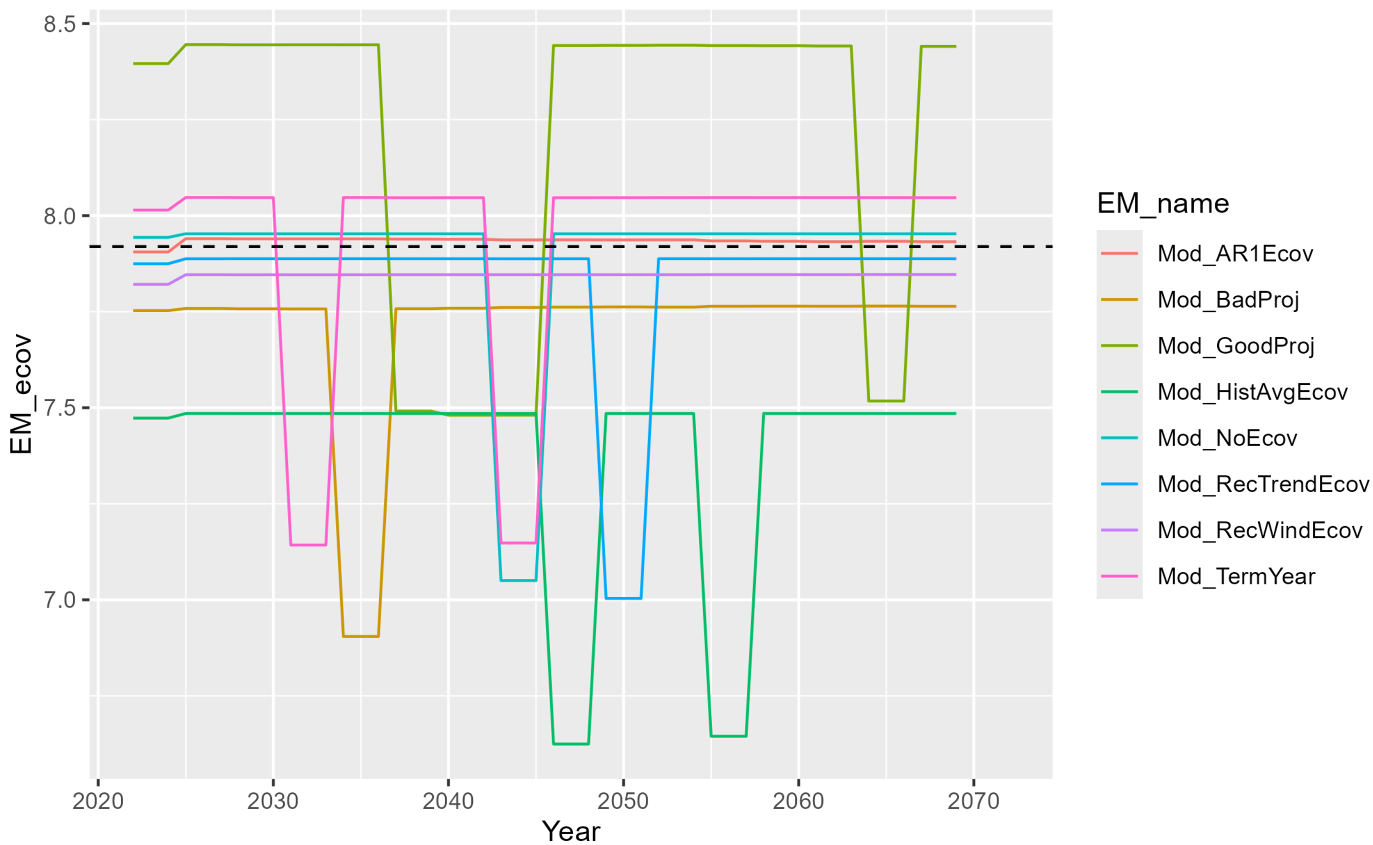
Table 1: Rho parameter estimates from each EM tested against each OM

EM_name	OM_1	OM_2	OM_3	OM_4
Mod_AR1Ecov	0.990	0.971	0.987	0.987
Mod_HIGHEcov	0.994	0.976	0.991	0.991
Mod_LOWEcov	0.920	0.922	0.939	0.945
Mod_MEDEcov	0.988	0.986	0.983	0.983
Mod_MEDLOWEcov	0.919	0.919	0.931	0.942
Mod_NoEcov	0.992	0.992	0.992	0.986

Table 2: Ecov parameter estimates, standard deviation, and rho for each EM for projection

EM_name	OM_ecov	mean_ecov	PercentChange	sd_ecov	rho
Mod_AR1Ecov	7.932	7.935	0.038	0.235	0.970
Mod_BadProj	7.764	7.707	-0.734	0.338	0.937
Mod_GoodProj	8.441	8.203	-2.820	0.340	0.981
Mod_HistAvgEcov	7.485	7.378	-1.430	0.348	0.863
Mod_RecTrendEcov	7.888	7.832	-0.710	0.691	0.863
Mod_RecWindEcov	7.847	7.845	-0.025	0.320	0.942
Mod_TermYear	8.047	7.932	-1.429	0.304	0.972

lowest parameter estimate was calculated from the EM which uses the historical average, which along with the recent trend projection, had the lowest logit rho value. Similar to the bias tests, the calculated parameter estimates at each assessment year did not appear to diverge from the estimate in the OM **??**. Figure **??** shows the catch advice, SSB, $\bar{F}$, and predicted catch from the OM of each projection test. Similar to the bias tests, these model outputs were very similar and all converged to similar estimations.

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This similarity is also captured in the model performance (figure 6), where the models have a similar performance, with the exception of the model without an ecological covariate and the model with a biased projction. The calculated SSB, ($\bar{F}$), and predicted catch from the terminal three years of the projection period show similar patterns of similarity across EMs (figure 7).

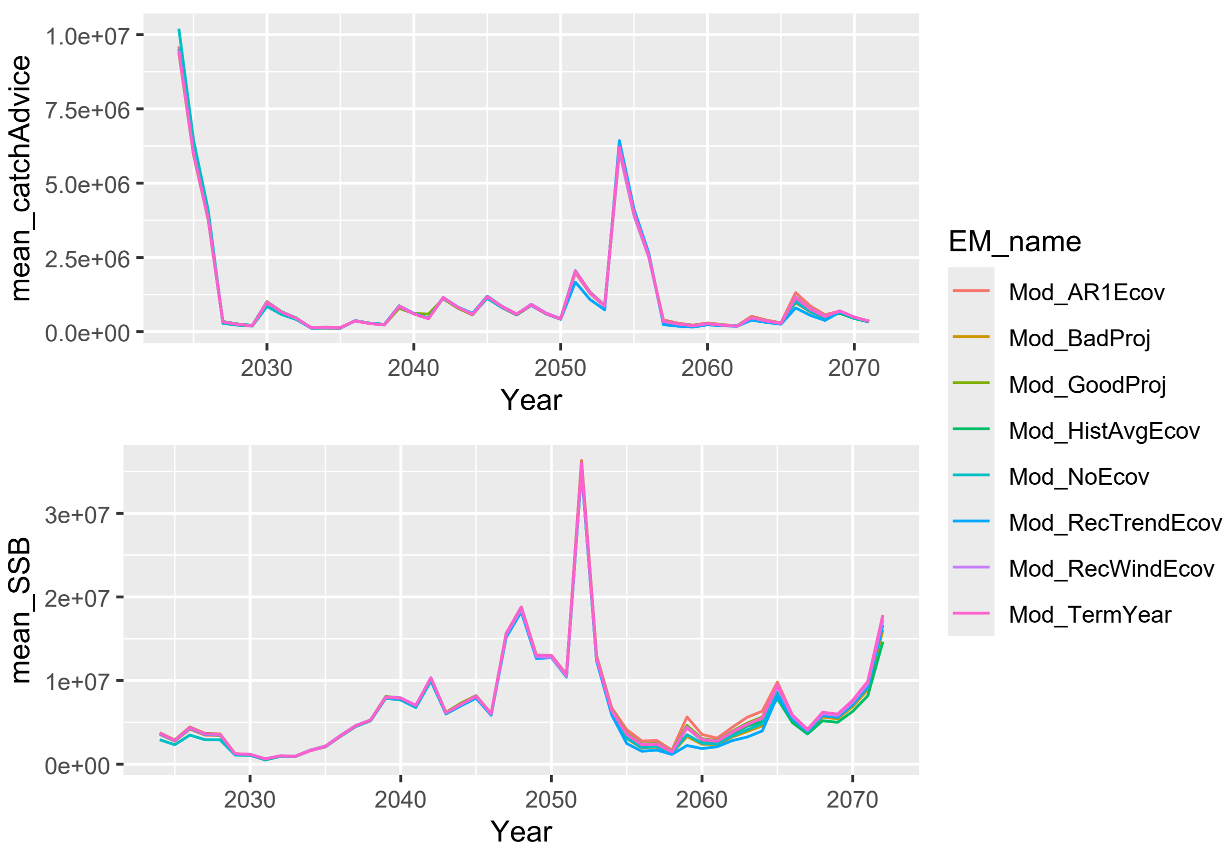

Figure 4: Catch advice (top), SSB (second), FBar (third), and predicted catch (bottom) calculated from each EM over the projection period

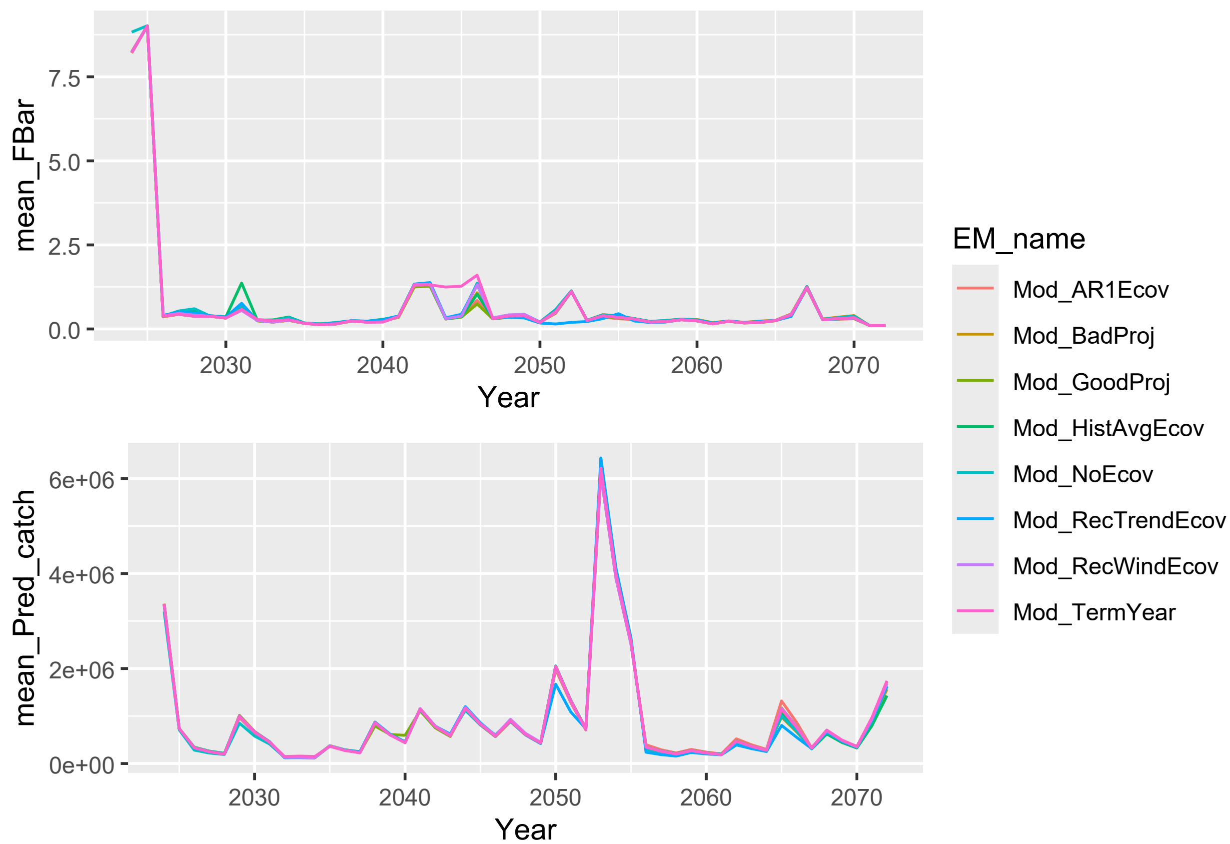

Figure 5: Catch advice (top), SSB (second), FBar (third), and predicted catch (bottom) calculated from each EM over the projection period

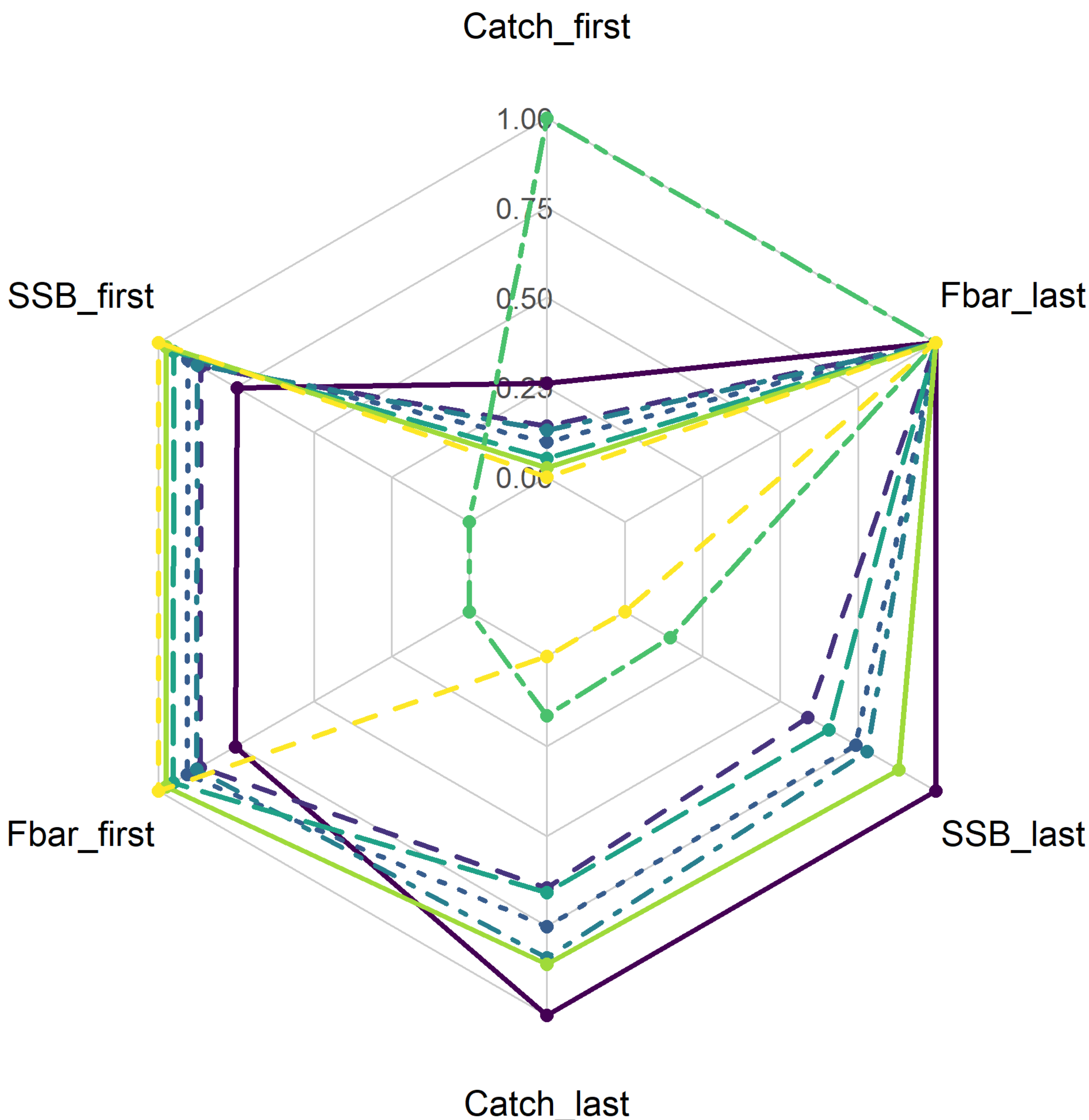

Figure 6: Relative performance of each EM

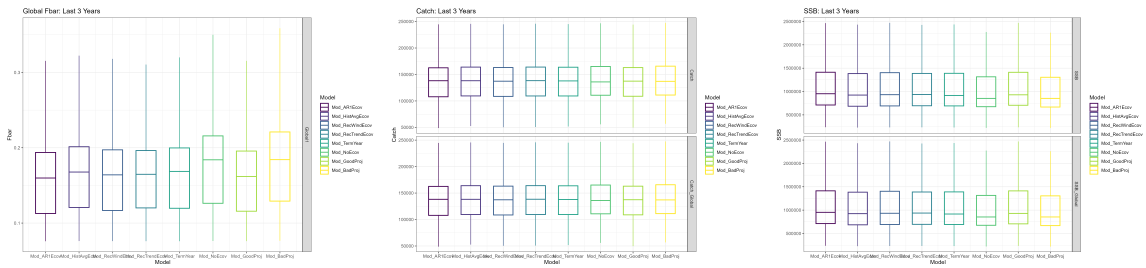

Figure 7: Fbar, SSB, and catch of terminal 3 years of MSE in projections test

5 DISCUSSION

The results from the bias test show that, while incorporating the ecological covariate helps model performance overall, the trajectory of the projection of temperature used in predicting future simulations can have a high or low bias, as they tended to underestimate the parameter from the 'true' value. However, the overall result had a limited impact on the SSB, $\bar{F}$, and subsequent catch. This is because regardless of the bias in the projection, the ecological parameter estimated at each assessment did not diverge away from the 'true' OM temperature regardless.

However, this test was more of a hypothetical exploration of how diverging temperature projections changes the outputs of the model. In reality, stock assessments use hindcast environmental information, so it is not realistic that an assessment would continue to use a temperature trajectory that diverges away from the real environmental conditions. Instead, stock assessments incorporate the most recent data and use this information to make predictions until the next assessment. This is why our projections test updated its temperature predictions with the previous three years of information. Similar to the bias tests, we found that incorporating the environmental covariate resulted in slightly different dynamics, but overall followed similar trajectories. The EM with the 'good' projection (i.e. one that followed that of the OM) performed the best, but other self-updating estimations of temperatures performed almost as well. This is in line with previous literature for yellowtail flounder, which has shown that there is little discernible difference between projections that used a continued estimated environmental process, ones that fixed the environmental process at recent 5 year average, and a third EM which used the observed environmental index (Britten et al., 2024). This could be the result of the type of link that the environmental covariate makes with the population. For Georges Bank Yellowtail, the environmental covariate has a lagged controlling effect on recruitment (CITE RT). Over time, the negative effects of high temperature could be masked by density-dependent mortality, or other factors that occur during the lag. Further, there may be a greater difference in model outputs if the ecov had a direct effect on catchability, mortality, behavioral, or metabolic processes; life history traits that can all be affected by environmental change (Stock & Miller, 2021), indicating a need for further research on projection choices for environmental covariates that affect other parts of life history. Catch advice is not very dependent on recruitment, and because this environmental link is also lagged, these combined factors could be masking the effect. Further, Haltuch & Punt (2011b) has shown that incorrectly including environmental covariates in the stock does not have a large effect on biomass reference points or relative errors of the model. A review by Punt et al. (2014a) has also shown that short and medium-term fisheries management simulations are generally not greatly improved by incorporating environmental covariates. This research served to test a longer-term simulation and found a similar result. However, if there is a high temporal contrast in the 'real world' or OM, incorporating the environmental covariate in the estimation model can improve model accuracy (Miller et al., 2023b).

Overall, we found that projection decisions had a limited effect on catch advice and fishery dynamics in an MSE framework, and this could be due to the nature of the environmental link to georges bank yellowtail flounder, indicating a need for further research on how this type of link will affect model dynamics over different projection decisions.

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